

PROJECT REQUIREMENT SPECIFICATION

Lean Hotel Management System (iOS MVP)

A React Native & FastAPI-Based Hotel Operations Platform

Field	Value
Document Type	Project Requirement Specification (PRS)
Version	1.0
Status	Final for Review
Date	August 30, 2026

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1. Project Overview

The proposed system is an iOS-focused Hotel Management application built with React Native (Expo) on the client and FastAPI with PostgreSQL on the backend, designed to run a single-property hotel's core operations within a lean, 7-day MVP delivery model.

The platform allows guests to search and book rooms directly from their phone, allows Front Desk staff to manage daily check-ins, check-outs, and room status in real time, and gives Admins full oversight of occupancy, room inventory, and staff accounts.

A role-based access model is implemented across three roles: Guest, Front Desk, and Admin. Admin holds the highest level of system access, including room inventory control and staff account management, while Front Desk is scoped to day-to-day, guest-facing operations.

2. Objectives

The system is designed to achieve the following objectives:

- Provide a mobile platform for guests to search, filter, and book hotel rooms.
- Prevent double-booking through database-enforced availability constraints.
- Give Front Desk staff a real-time view of today's arrivals, departures, and room status.
- Allow Front Desk staff to update room status as guests check in and check out.
- Give Admins a property-wide dashboard of occupancy and operational performance.
- Allow Admins to manage room inventory (add, edit, deactivate rooms).
- Allow Admins to create and deactivate Front Desk staff accounts.
- Provide an AI-powered concierge to answer guest questions about hotel policies and amenities.
- Maintain a clear, role-based separation of access across Guest, Front Desk, and Admin.
- Deliver a stable, demoable MVP within a 7-day development window.

3. User Roles

The system defines three primary user roles, each with a distinct scope of responsibility and access.

3.1 Guest

Guests will be able to:

- Register and log in to the application.
- Search available rooms by check-in and check-out date.
- View room type, capacity, price per night, and photos (if provided).
- Select a room and confirm a booking.
- View their own upcoming and past bookings.
- Cancel a booking prior to check-in, subject to hotel policy.
- Chat with the AI Concierge about amenities, policies, and stay-related questions.

During booking, the system will validate:

- Selected date range against current room availability.
- Prevention of overlapping reservations for the same room.
- Correct price calculation across the full length of stay.

3.2 Front Desk

Front Desk staff manage day-to-day guest operations, including:

- Log in with Front Desk credentials.
- View today's arrivals, departures, and current room status grid.
- Check a guest in (moves a booking from Confirmed to Checked-In).
- Check a guest out (moves a booking from Checked-In to Checked-Out).
- Update room status (Available, Occupied, Cleaning, Maintenance).
- View current occupancy at a glance to answer walk-in guest questions.
- View booking details for any guest currently on the property.

3.3 Admin

The Admin holds the highest level of system access, including:

- Log in with Admin credentials.
- View the full Admin Dashboard: occupancy rate, booking volume, and room inventory status.
- Add, edit, or deactivate rooms (type, capacity, nightly price, description).
- Create and deactivate Front Desk staff accounts.
- View property-wide booking history and occupancy trends.

- Review audit logs of key system actions.
- Configure system-level settings (e.g., hotel policy text used by the AI Concierge).

4. Authentication and User Management

The system will provide secure authentication with the following features:

- Guest self-registration and login using email and password.
- Front Desk and Admin accounts are provisioned by an Admin — not self-registered — to keep staff access controlled.
- JWT-based session tokens with a role claim (GUEST, FRONT_DESK, or ADMIN).
- Passwords hashed with bcrypt; plaintext passwords are never stored or logged.
- Role-based authorization enforced on every protected API route.
- Secure logout that invalidates the client-side session token.

5. Room & Inventory Management

Admins manage the property's room inventory through a dedicated module. Each room record includes the following attributes:

Field	Example Value
Room Number	204
Room Type	Deluxe Double
Capacity	2 Adults
Price per Night	PKR 12,500
Status	Available
Description	City-view room with king bed and workspace

Admins can add new rooms, edit existing room details, and deactivate rooms that should no longer appear in the Guest catalog (e.g., rooms undergoing long-term renovation).

6. Room Catalog and Availability Search

Guests browse rooms through a catalog screen that supports:

- Filtering by check-in and check-out date.
- Filtering by room type and guest capacity.

- Displaying only rooms with full availability across the entire selected date range.
- Showing price per night and total price for the selected stay length.

Availability is computed server-side against the booking data — the client never determines availability locally, which keeps the catalog consistent even under concurrent bookings.

7. Booking Engine

The booking engine is the core transactional component of the system. When a guest confirms a booking, the system:

- Validates the selected room and date range are still available at the moment of submission.
- Writes one availability record per night of the stay inside a single database transaction.
- Relies on a uniqueness constraint on (room, date) so that a conflicting booking attempt fails the entire transaction rather than partially succeeding.
- Returns a confirmation with a unique booking reference on success, or a clear conflict response if the room was booked by another guest first.

This design guarantees zero double-bookings even when two guests attempt to book the same room and dates at the same moment, without requiring complex manual locking logic.

8. Booking Status Lifecycle

A booking progresses through the following statuses:

Status	Description
Pending	Booking request submitted; awaiting final confirmation.
Confirmed	Booking has been validated and the room is reserved for the stay dates.
Checked-In	Front Desk has checked the guest into the room.
Checked-Out	Guest has completed their stay and checked out.
Cancelled	Booking was cancelled by the guest or Front Desk prior to check-in.

9. Front Desk Operations Module

The Front Desk module is the primary working screen for daily hotel operations. It displays:

- Today's expected arrivals with guest name, room, and booking reference.
- Today's expected departures.
- A live grid of all rooms and their current status.
- Check-in and check-out actions, each of which updates both the booking status and the room status in a single action.

Checking a guest out automatically transitions the room to Cleaning status, so Front Desk staff always have an accurate, real-time view of which rooms are ready for the next guest.

10. Room Status Management

Each room carries one of four statuses at any given time:

Status	Description
Available	Room is clean and ready to be booked or occupied.
Occupied	A guest currently holds the room under an active stay.
Cleaning	Room has been vacated and is awaiting housekeeping.
Maintenance	Room is temporarily unavailable due to repairs or upkeep.

Front Desk staff can transition a room between these statuses; Admins retain the same capability plus the ability to deactivate a room from inventory entirely.

11. Admin Dashboard

The Admin Dashboard provides property-wide KPI cards, including:

- Total Rooms.
- Occupied Rooms.
- Today's Check-ins.
- Occupancy Rate (percentage).
- Total active and upcoming bookings.

KPI values are computed via a single aggregate query and refreshed on a short polling interval, giving Admins a near-real-time view without requiring a persistent websocket connection.

12. Reports and Booking History

Admins can view a chronological list of all bookings across the property, filterable by date range and booking status, to review recent activity and spot trends in occupancy. Exportable reports (PDF/Excel) are a natural v2.0 extension once the core operational flows are stable.

13. AI Guest Concierge

Guests can open a chat screen to ask questions about their stay. The concierge is powered by the OpenAI gpt-4o-mini model, with a static hotel-policy document (Wi-Fi details, pool hours, breakfast times, check-in/checkout times, and cancellation policy) injected into the system prompt.

Guest Question	Concierge Answer Source
What time is breakfast served?	Static hotel policy text (breakfast hours)
Is there a pool?	Static hotel policy text (amenities list)
What is your cancellation policy?	Static hotel policy text (cancellation policy)

This approach avoids the need for a retrieval or vector-search system within the 7-day timeline while still giving guests fast, accurate answers to the most common questions.

14. Concierge Security and Data Handling

- The OpenAI API key is stored server-side only and is never exposed to the client application.
- All concierge requests are proxied through the FastAPI backend, which attaches the hotel policy context before forwarding the guest's question.
- Guest chat messages are not used to answer other guests' questions and are not retained beyond what is needed to render the conversation.

15. Notifications (Future Scope)

Push notifications via Apple Push Notification service (APNs) are out of scope for the 7-day MVP. The following notification events are documented here for v2.0 planning:

- Booking confirmed.
- Check-in reminder.
- Checkout reminder.
- Booking cancelled.

16. Audit Logs

The system maintains a lightweight audit trail for key actions, such as:

- A booking was created or cancelled.
- A room's status was changed, and by whom.
- An Admin created, edited, or deactivated a room.
- An Admin created or deactivated a Front Desk staff account.

Each audit log entry records the acting user, the action taken, the affected record, and a timestamp.

17. Non-Functional Requirements

Security

- Secure authentication with hashed passwords.
- Role-based authorization enforced on every protected endpoint.
- Input validation on all API requests via Pydantic schemas.
- TLS/HTTPS enforced for all client-server traffic.

Performance

- Room catalog queries target under 200ms.
- Booking confirmations target under 500ms.
- The system should support multiple guests searching and booking simultaneously without data inconsistency.

Reliability

- Zero double-bookings, enforced at the database layer rather than in application code alone.
- Booking transactions are atomic — a booking either fully succeeds or fully fails, with no partial state.

Usability

- Simple, guided booking flow for guests.
- A clear, glanceable status grid for Front Desk staff.
- Dashboard KPIs presented in plain, non-technical language for Admins.

18. Security and Compliance

- Role-Based Access Control (RBAC) across Guest, Front Desk, and Admin, enforced via a shared authorization dependency reused across all protected routes.
- Front Desk routes reject Guest tokens; Admin routes reject both Guest and Front Desk tokens.
- Secrets (database credentials, JWT signing key, OpenAI API key) are stored as environment variables and never committed to source control.
- No payment or government-ID data is collected in the MVP, limiting the system's compliance surface area.

19. Proposed Technology Stack

Layer	Technology	Purpose
Mobile Application	React Native (Expo)	Guest, Front Desk, and Admin iOS application.
Backend	FastAPI (Python 3.11)	REST APIs, authentication, booking logic, concierge proxy.
Database	PostgreSQL (Supabase / Neon)	Rooms, bookings, users, roles, audit logs.
ORM / Migrations	SQLAlchemy 2.0 + Alembic	Schema management and data access.
Authentication	JWT (python-jose) + bcrypt	Token-based auth and password hashing.
AI Concierge	OpenAI gpt-4o-mini	Natural-language answers to guest questions.
State Management	Zustand	Lightweight client-side state for the mobile app.
Navigation	expo-router	Screen navigation within the Expo app.

20. Major System Modules

The complete system can be organized into the following functional modules:

- Authentication Module
- Guest Management
- Front Desk Management
- Admin Management
- Room Inventory Module
- Room Catalog & Search Module
- Booking Engine Module
- Booking Lifecycle Module
- Room Status Module
- Admin Dashboard Module
- Reports & Booking History Module
- AI Concierge Module
- Audit Log Module
- Notification Module (Future)
- System Settings Module

21. Overall Workflow

The end-to-end process flow across roles is summarized below:

Stage	Role	Action
1	Admin	Set up and maintain room inventory (type, capacity, price, status).
2	Admin	Create and manage Front Desk staff accounts.
3	Guest	Register/log in, search rooms by date range, and view availability.
4	Guest	Select a room and confirm a booking.
5	System	Booking engine validates availability and prevents overlapping reservations.
6	Guest	Use the AI Concierge to ask stay-related questions.
7	Front Desk	View today's arrivals/departures and check guests in.

Stage	Role	Action
8	Front Desk	Check guests out; room status automatically moves to Cleaning.
9	Front Desk	Update room status back to Available once cleaning is complete.
10	Admin	Review the dashboard, booking history, and audit logs.

22. Out of Scope for MVP (v1.0)

To guarantee delivery within the 7-day timeline, the following are explicitly deferred to v2.0:

- Payment gateway processing (Stripe/Apple Pay) — simulated via instant confirmation for the MVP.
- Native hardware features (NFC digital room keys, CoreBluetooth door locks).
- Multi-hotel enterprise support (single-property system model only).
- Push notifications via Apple Push Notification service (APNs).
- Passport/ID image OCR parsing.
- Retrieval-augmented (RAG) knowledge base for the concierge — static hotel policy text only.
- Exportable PDF/Excel reports.

23. Release Metrics and Verification

- Zero Overbooking Error Rate: 100% success rate on automated concurrency tests attempting double-bookings on identical room dates.
- End-to-End Flow Completion: successful execution of the primary guest journey (Registration → Room Search → Booking Confirmation → Concierge Chat) on a physical iOS device within 60 seconds.
- Deployment Stability: no critical API server crashes during the 7-day sprint review demo.

24. Key Requirement

Core Success Factor

The most important part of this project is not the booking screen in isolation, but the complete guest-to-stay lifecycle as a whole.

The system must combine the mobile application, backend booking engine, role-based Front Desk and Admin operations, and the AI Concierge into one coherent, demoable Hotel Management platform — not a set of disconnected screens.